

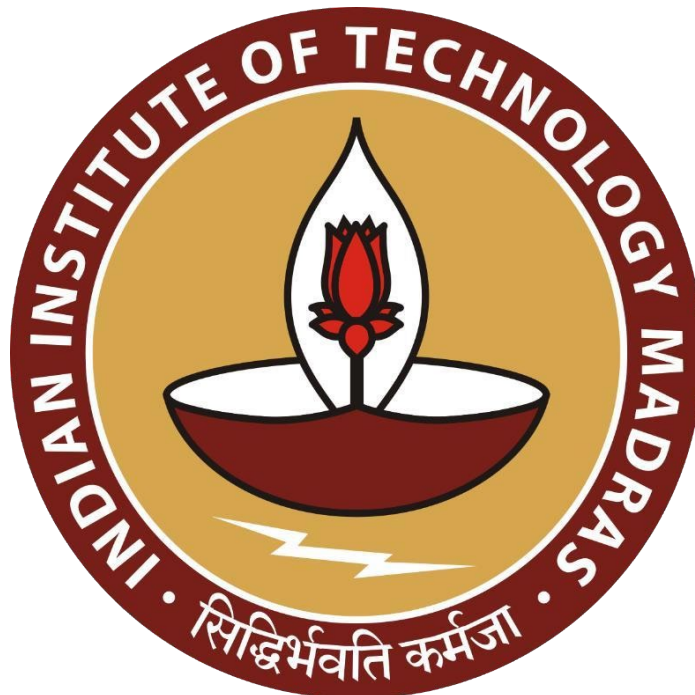
Optimal Inventory Management and Sales Forecasting for Janananma Pharmacy

Mid-Term Report for the BDM Capstone Project

Submitted by

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Contents

1	Executive Summary and Title	2
2	Proof of Originality of Data	3
3	Meta Data and Descriptive Statistics	4
4	Detailed Explanation of Analysis Process/Method	7
5	Results and Findings	9

1. Executive Summary and Title

Title: Optimal Inventory Management and Sales Forecasting for Janananma Pharmacy

Janananma is a B2C leading healthcare provider committed to deliver quality pharmaceutical products and services to customers. Established in 2018, the pharmacy has grown to become a trusted name in Palakkad district Kerala. The pharmacy offers a comprehensive range of prescription and over-the-counter medications, wellness products, and surgical products.

In a highly competitive and regulated healthcare market, Janananma facing key operational challenges that impact its efficiency and profitability. These challenges include inventory management issues, where maintaining optimal stock levels has been difficult, sometimes they face stockout issues and sometimes more expense for unnecessary stocks; expiry management, where products nearing their expiry dates are not being utilized or disposed of efficiently and keep more stocks than required also leads to expire the products; and brand diversity, which complicates procurement and sales strategies. Addressing these problems is essential to reducing financial losses, enhancing customer satisfaction, and improving operational efficiency.

This project focuses on analysing the sales, purchase, expiry, and inventory data over the past six months, based on 19,021 sales records and 3,447 purchase records from 01/10/2023 to 31/03/2024. The approach involves data cleansing, visualization, and deriving actionable insights. By analysing these data, it is possible to create a better inventory management by identifying high-demand products, using sales data that should be prioritized in inventory, minimizing stockouts and overstocking. And identifying the product with low and high expiry duration will help to enhance strategies for handling product expiries. Analysis on the doctors and the product brand help to point out the most demanded product brand as well, this will reduce the issue on brand diversity.

By using advanced analytical tools and predictive analytics, the project aims to forecast sales which leads to better inventory management, prevent stockouts. The expected outcomes include improved stock management practices, enhanced strategies for handling product expiries, and optimized brand diversity decisions. The solutions provided aim to help the pharmacy tackle these challenges, enabling it to make more informed business decisions and achieve operational excellence.

2. Proof of originality of the Data

Details of the Business

Name of the Business : Janananma medicals and surgicals
Name of the Owner : Muhammed Ajmal
Place of the Business : Mannarkkad, Palakkad, Kerala
Starting Year of the Business : 2018
Number of Employees : 5



Figure 1: Janananma Medicals and Surgicals



Figure 2: Myself with Proprietor


JANANANMA
MEDICALS & SURGICALS
HOSPITAL JUNCTION, MANNARKKAD, PALAKKAD
Mob : 8714 222 170, 9037 114 184
DL NO: KL-PKD-108083
KL-PKD-108084

23-May-2024

TO WHOM IT MAY CONCERN

This is to certify that I Muhammed Ajmal proprietor of Janananma Medicals and surgicals, Mannarkkad, Palakkad have provided business data for the duration year 2023 October – 2024 March to Nihita Pallath for the purpose of academic project of Business Data Management in IIT Madras.

The data provided by me is true to my knowledge and this can be used only for academic purpose.

Mannarkkad


Janananma medicals and surgicals

JANANANMA MEDICALS & SURGICALS
Door No: 10615,
HOSPITAL JUNCTION, MANNARKKAD,
PALAKKAD DIST. PALAKKAD, PIN-686 002

Figure 3: Authorization letter from the Organization

Link to the Meeting with the Proprietor:

https://drive.google.com/file/d/1oeQLO1fTI_XAiL0y4by8mJlr1k3O59z/view?usp=sharing

Link to the Data Collected:

<https://drive.google.com/drive/folders/1vlyHwUu2b6hmubSS3Iuu-hz6noIasicW?usp=sharing>

3. Metadata and Descriptive Statistics

The data collected for the duration of past 6 months [01/10/2023 - 31/03/2024] from Janananma Medicals and Surgicals. Since the data is stored in a software application, it provides separate data tables for purchase, sales, expiry and inventory. It was an unorganized data. The purchase data is in the form of product-wise purchase statement and sales are in the form of bill statement.

The data sets contain the following columns:

- Sales Data: Each record represents each sale to the customer.
['Patient Name', 'Doctor Name', 'Date', 'Product', 'Batch No.', 'Expiry Date', 'Qty', 'Rate', 'Discount', 'Amount', 'Tax', 'Year-Month']
- Purchase Data: Each record represents a product purchased by the pharmacy from the companies
['Product', 'A/cDate', 'Bill No', 'BillDat', 'Supplier', 'BatchNo', 'Expiry', 'Qty', 'Amount', 'TaxAmt', 'Total']
- Inventory Data: It contains the stock of each product on 01/10/2023 (Initial stock)
['Product', 'Qty', 'Rate', 'OrgVal']
- Expiry Data: Expiry date of each product and its details.
['Product', 'BatchNo', 'Expiry', 'Qty', 'MRP']

JANANANMA MEDICALS & SURGICALS									
HSPTL JUNC , NEAR KPM HOTEL, MANNARKKAD, Whatsapp : 8129395919									
Bill Statement									
From : 01/10/2023 To : 31/03/2024					page no 1				
bill n	17895	mode	CASH	date	01/10/2023				
patien	VARGHESE								
product	batchno	expiry	qty	rate	dis%	amt	tax%		
GLUVILDA M 500	GLD23013M	03/2025	30	9.20	10.00	248.40	12.00		
CLOPIVAS AP-150	EC0313	03/2025	15	6.33	10.00	85.50	12.00		
VOGLET M0.3 TAB	v2275601	10/2024	30	8.00	10.00	216.00	12.00		
PROLOMET XL-25 TAB	GTE1096A	04/2025	30	4.70	10.00	127.01	12.00		
MONIT GTN 2.6 TAB	N2301333	04/2025	30	7.53	10.00	203.40	12.00		
PANTOP.D	SPD230449	03/2025	15	10.65	8.00	146.97	12.00		
BIOPRIDE 2 TAB	ajl23008	01/2025	30	5.42	10.00	146.34	12.00		
DIGEPLEX-T	DTSC004	03/2024	20	14.88	7.00	276.77	12.00		
ROSUMAC-5mg tab	KRK2301A	02/2026	10	8.87	10.00	79.80	12.00		
D-PROTIEN 200GM VANIL	DPVA22080	07/2024	1	280.00	3.50	270.20	12.00		
MURIVENNA	665	07/2024	1	30.00	0.00	30.00	5.00		
						total :	1830.39		
						tax :	194.34		
						net amt :	1830.39		

Figure 4: Snapshot of raw sales data

Patient Na	Doctor Name	Date	Product	Batch No.	Expiry Date	Qty	Rate	Discount	Amount	Tax	Total	YearMonth
AYSHA	RANJITH	01-10-2023	ACILOC 150 TAB 30,	LD22304	31-10-2024	4	1.3647	4	5.24	12	5.895056	2023-10
ASHNA	189	01-10-2023	AEROCORT ROTOC	SA30955	31-10-2024	1	114.35	10	102.92	12	116.642	2023-10
JANU	AJAY K GOPA	01-10-2023	ANGICAM BETA TAB	agb2323	30-04-2025	2	2.5333	2	4.97	12	5.577992	2023-10
YOUSUF K	ANISH AHAM	01-10-2023	APIDRA VIAL	2F004A	31-10-2024	1	1534	14	1319.24	5	1395.94	2023-10
RENU	ABDUL RIYAL	01-10-2023	BABY DIAPER RASH	8.52E+08	31-08-2025	2	80	0	160	12	179.2	2023-10
RENU	ABDUL RIYAL	01-10-2023	BABY DIAPER RASH	282	31-03-2026	1	80	0	80	12	89.6	2023-10
		01-10-2023	BAND-AID	C647	31-03-2024	7	2	0	14	12	15.68	2023-10
SHAMNA	NIBIT PURKA	01-10-2023	BETADINE-OINT	PD0813	31-03-2025	1	145.14	0	145.14	12	162.5568	2023-10
MUHAMME	RANJITH	01-10-2023	BILASURE 20	AXA0025	31-01-2025	7	13.9	3	94.38	12	106.056	2023-10
VARGHESE	SURESH	01-10-2023	BIOPRIDE 2 TAB	ajl23008	31-01-2025	30	5.42	10	146.34	12	165.852	2023-10
VASUDEVA	RAISON P CH	01-10-2023	BIOTOR CV 10	LE223052	30-04-2025	5	10.45	3	50.68	12	56.95	2023-10
SARATH	SURESH	01-10-2023	BRIV+50	BRITB2023	30-09-2024	30	15.4	7	429.66	12	485.1	2023-10

Figure 5: Snapshot of cleaned sales data

Where,

- Qty: sold quantity

- Rate: Rate for each product
- Amount: $\text{Quantity} \times \text{Rate} - \text{Discount} \times (\text{Quantity} \times \text{Rate})$
- Total: $\text{Amount} + (\text{Tax}/100) \times (\text{Quantity} \times \text{Rate})$

The first step in data cleaning process was to organize the obtained sales and purchase statement data into a table format. A vba macro code in excel used to convert the sales and purchase data into structured table format. The cleaned data contains 50890 rows. To ensure a focused and meaningful analysis, the data was filtered to retain only the top-selling products. A threshold of 4,000 sales per month was set as the benchmark, with products falling below this threshold excluded from further analysis. This filtering process was implemented to eliminate less impactful data points and concentrate on high-performing products that contribute significantly to the pharmacy's revenue and inventory management.

Descriptive statistics for sales data:

Python pandas used as the tool to find the descriptive statistics – `data.describe()`

	Qty	Rate	Discount	Amount	Tax	UnitCost	Total
count	3164.000000	3164.000000	3164.000000	3164.000000	3164.0	3164.000000	3164.000000
mean	30.202592	2.442102	8.536504	54.536220	12.0	3.723653	66.536220
std	48.290344	1.567276	2.495176	73.119064	0.0	2.441686	73.119064
min	1.000000	0.245200	0.000000	1.110000	12.0	0.420667	13.110000
25%	7.000000	1.397000	9.000000	12.100000	12.0	1.862667	24.100000
50%	12.000000	1.625000	10.000000	26.320000	12.0	3.119286	38.320000
75%	30.000000	3.214500	10.000000	65.780000	12.0	4.566000	77.780000
max	450.000000	7.228600	15.000000	493.880000	12.0	15.840000	505.880000

Figure 6: Descriptive statistics of sales data

The most important columns in this descriptive statistics are Qty which represents the sold quantity and Total which represents the total selling price. The key insights from this sales data descriptive statistics are,

- Sales range from 1 unit to 450 units, with a high standard deviation (48.29), indicating variability in transaction sizes. The large variation in transaction sizes suggests that both small and bulk purchases are common.
- With maximum totals as high as 505.88 as the total amount for a selling indicates bulk purchases or expensive product bundles, it significantly contributes to revenue.

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Purchase Statement - Productwise										
From : 01/10/2023 To : 31/03/2024										Page No 1
SI No	A/cDate	Bill No	BillDat	Supplier	BatchNo	Expiry	Qty	Amount	TaxAmt	Total
4-QUIN LOT EYE DROP										
1740	24/11/2023	21590	#####	ATTUPURAM DRUGS	23156	08/2025	1	138.57	16.63	155.20
							1	138.57	16.63	155.20
4-QUIN-DROPS										
1740	24/11/2023	21590	#####	ATTUPURAM DRUGS	A-3048	07/2025	2	216.17	25.94	242.11
							2	216.17	25.94	242.11

Figure 7: Snapshot of raw purchase data

Product	A/cDate	Bill No	BillDat	Supplier	BatchNo	Expiry	Qty	Amount	TaxAmt	Total
CIFRAN 25	09-04-2024	300956	08-04-2024	NEDSONS	SXE2171A	31-10-2026	170	285.76	34.29	320.05
CIFRAN 25	15-04-2024	237	12-04-2024	YEMMARK	SXE2172A	31-10-2026	70	117.67	14.12	131.79
CIFRAN 25	17-04-2024	1617	15-04-2024	KURIKKAL	SXE1803A	31-07-2026	70	118.89	14.27	133.16
CIFRAN 25	17-04-2024	1617	15-04-2024	KURIKKAL	SXE0257A	31-01-2026	10	15.21	1.83	17.04
CIFRAN 25	17-04-2024	1617	15-04-2024	KURIKKAL	SXD2322A	30-09-2025	20	35.69	4.28	39.97
CIFRAN 25	17-04-2024	4623	16-04-2024	KIMS MEDI	SXE2172A	31-10-2026	350	594.46	71.34	665.8

Figure 8: Snapshot of cleaned purchase data

Descriptive statistics for purchase data:

	Qty	Amount	TaxAmt	Total	UnitCost
count	46.000000	46.000000	46.000000	46.000000	46.000000
mean	419.913043	614.338696	73.720217	688.058913	1.911037
std	457.324384	572.347990	68.681539	641.029529	1.345334
min	10.000000	0.000000	0.000000	0.000000	0.000000
25%	100.000000	194.310000	23.315000	217.625000	1.167040
50%	325.000000	479.200000	57.505000	536.705000	1.594000
75%	500.000000	735.360000	88.240000	823.600000	1.979350
max	1800.000000	2081.950000	249.830000	2331.780000	5.551857

Figure 9: Snapshot of cleaned purchase data

- The standard deviation of Qty 457.32 indicates high fluctuations in purchase quantity, indicates some medicines stocks in a bulk quantity. So those are taking up more space in inventory. Checking sales of those products can help to manage the storage cost and expiry as well.
- The mean (419.91 units) is higher than the median (325 units), suggesting that there are some very large purchases, those can be outliers.
- The highest quantity purchased in a month is 1800 units, which is significantly higher than the median (325), which indicates some medicines frequently restocked in bulk.
- In Total Min: ₹0.00 are possibly promotional items or free samples
- From Figure 10, products with the highest total selling price and quantity values indicate strong demand. So, CIFRAN 250 TAB has more demand in this.

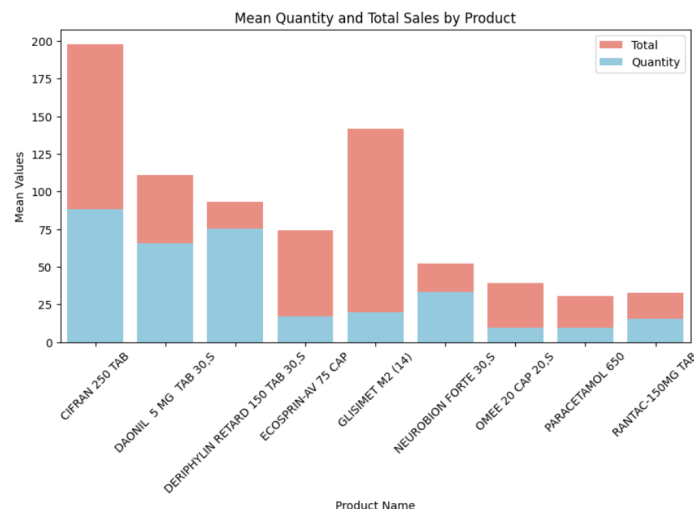


Figure 10: Bar chart for Mean of Quantity and Total sales for each filtered product

4. Detailed Explanation of Analysis Process/Method

Data Cleaning and Preprocessing: The analysis process began with data collection and transformation where two primary datasets were used: sales data, representing the medicines purchased by customers from the pharmacy, and purchase data, capturing the procurement activities conducted by pharmacy personnel. These datasets were initially in sales receipt and purchase statement models. To standardize and structure the information for analysis, Excel vba macro (Refer Figure 11) employed to convert these models into well-structured Sales and Purchase tables.

```
(General)
' Set the headers in the new sheet
newWs.Cells(1, 1).Value = "Patient Name"
newWs.Cells(1, 2).Value = "Doctor Name"
newWs.Cells(1, 3).Value = "Date"
newWs.Cells(1, 4).Value = "Product"
newWs.Cells(1, 5).Value = "Batch No."
newWs.Cells(1, 6).Value = "Expiry Date"
newWs.Cells(1, 7).Value = "Qty"
newWs.Cells(1, 8).Value = "Amount"

' Initialize row for new data
newRow = 2

' Find the last row of data in the original sheet
lastRow = ws.Cells(ws.Rows.Count, "A").End(Up).Row

' Loop through all rows in the original sheet
For i = 1 To lastRow
    ' Check for patient name
    If ws.Cells(i, 1).Value = "Patient" Then
        patientName = ws.Cells(i, 2).Value
    End If

    ' Check for doctor name
    If ws.Cells(i, 7).Value = "Dr. " Then
        doctorName = ws.Cells(i, 8).Value
    End If

    ' Check for date
    If ws.Cells(i, 7).Value = "date" Then
        dateValue = ws.Cells(i, 8).Value
    End If

    ' Check for product details
    If ws.Cells(i, 1).Value = "Product" Then
        j = 2
        Do While ws.Cells(j, 1).Value <> "" And ws.Cells(j, 1).Value <> "bill n"
            ' Transfer data to new sheet
            newWs.Cells(newRow, 1).Value = patientName
            newWs.Cells(newRow, 2).Value = doctorName
            newWs.Cells(newRow, 3).Value = dateValue
            newWs.Cells(newRow, 4).Value = ws.Cells(j, 1).Value
            newWs.Cells(newRow, 5).Value = ws.Cells(j, 3).Value ' Batch No. is in the third column from product
            newWs.Cells(newRow, 6).Value = ws.Cells(j, 4).Value ' Expiry Date is in the fourth column from product
            newWs.Cells(newRow, 7).Value = ws.Cells(j, 5).Value ' Qty is in the fifth column from product
            newWs.Cells(newRow, 8).Value = ws.Cells(j, 6).Value ' Qty is in the fifth column from product
            newRow = newRow + 1
            j = j + 1
        Loop
    End If
    Next i

' Display a message box indicating completion
MsgBox "Activity completed", vbInformation
End Sub
```

Figure 11: vba macro-code for sales data

Data cleaning and preprocessing followed as crucial steps to enhance data quality and ensure accurate analysis. Unnecessary data columns were removed to reduce noise and focus on meaningful attributes like patient name, bill number, A/C Date, Tax, Discount, Price. Missing values, though minimal, were addressed by inputting the most frequent values for categorical columns except the doctor's name, for the doctor column I used unique doctor names for each missing values and the mean for numerical columns. Furthermore, a threshold of 4,000 units per month was set to filter out low-selling products, ensuring that the analysis focused on high-demand products with significant business impact.

Stock Replenishment Analysis: To optimize inventory management and address stock shortage issues, a

comprehensive stock replenishment analysis was conducted. It's also calculated the Reorder Point and Economic order quantity for this.

1. Reorder Point (ROP) - The first step involved calculating the lead time for each order from the purchase data, followed by determining the average lead time per product, rounding up to the nearest whole number as it represents days. Safety stock for each product was then computed using the formula:

$$\text{Safety Stock} = (\text{Maximum Daily Sales} \times \text{Maximum Lead Time}) - (\text{Average Daily Sales} \times \text{Average Lead Time}).$$

The average safety stock value was derived for each product to account for demand fluctuations and supply chain uncertainties. Figure 12 gives the graphical representation of the medicines considered.

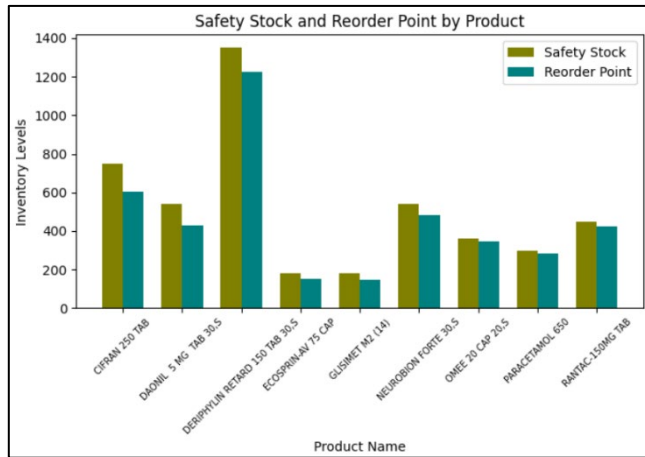


Figure 12: Safety stock and reorder point for each product

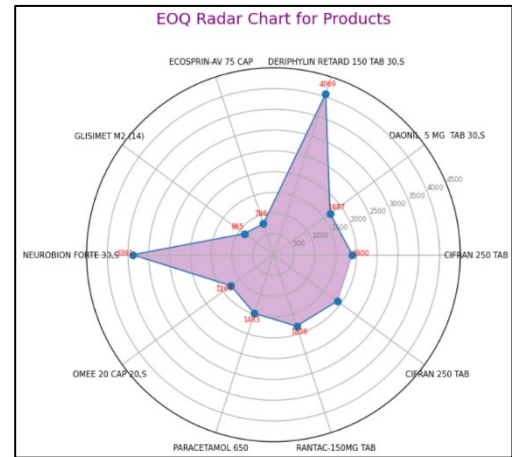


Figure 13: Radar graph of EOQ for each product

- For all the products having very less difference in safety stock and reorder point . It is because the lead time for all products is less. So, there is no need to spend storage waste by keeping more stock for more days.
- Products such as DERIPHYLIN RETARD 150 TAB and CIFRAN 250 TAB demand careful monitoring due to their high reorder points and safety stocks.
- Products with lower safety stocks, such as ECOSPRIN-AV 75 CAP, indicate stable demand and lead time reliability.

2. Economic Order Quantity (EOQ) – It is to determine the ideal order size that minimizes total inventory costs, which include both ordering and holding costs. For the ordering cost, a fixed cost of ₹30 per order was considered, based on procurement from Palakkad suppliers with standard transportation charges. The holding cost incorporates expenses such as storage, insurance, depreciation, expiry risks, and opportunity costs. Due to limited data on these specific costs, a general industry practice of 25% of the medicine's unit purchase cost was applied. For calculating EOQ the formula given below was used,

$$EOQ = \sqrt{\frac{2DS}{H}}$$

Where, D – Annual Demand, S – Ordering cost per order, H – Holding cost per unit

Figure 13 gives the radar plot of optimum EOQ for each product under analysis, From this:

- DERIPHYLIN RETARD 150 TAB (4069) and NEUROBION FORTE 30 (3391) has the highest EOQ, indicating that it requires a larger order size. It indicates the demand of the product.
- OMEE 20 CAP (1267), PARACETAMOL 650 (1483), and GLISIMET M2 (865) have lower EOQ values, meaning smaller batches are optimal due to low demand or reduced holding costs.

Sales Forecasting: The prediction on future sales trends was carried out based on historical sales data, time-series forecasting was utilized due to the sequential and temporal nature of the data. This approach is highly effective in identifying seasonal trends, fluctuations, and long-term patterns, which are critical for managing pharmacy inventory efficiently. As its observed, the trend of the sales per month for each product, almost all products keeps a stable sales for all month. Considering these characteristics of the sales data, the ARIMA (Autoregressive Integrated Moving Average) model was selected for the forecasting process. ARIMA is particularly suitable for stationary data that can be made stationary through differencing, effectively modeling both trend and noise components. The forecasts offer valuable guidance for inventory planning and procurement. Products with consistent sales trends benefit from accurate forecasts, while high-demand products with fluctuating trends require closer monitoring to adjust stock levels dynamically.

5. Results and Findings

In the original sales data, the initial analysis of the sales distribution revealed that out of 3,523 products, 2,286 products had total sales quantities under 100 units over the past six months. The histogram visualization, with a logarithmic scale, clearly highlighted this skewed distribution where most of the products experienced minimal sales activity. Such a significant concentration of low-selling products underscored the need for setting a sales threshold to filter out these low-demand items from further analysis.

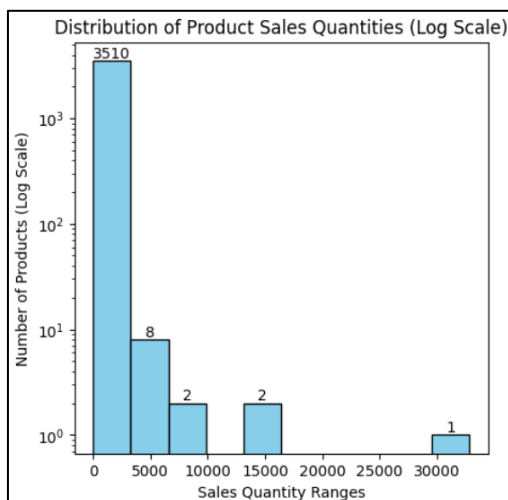


Figure 14: Histogram for distribution of sales quantities

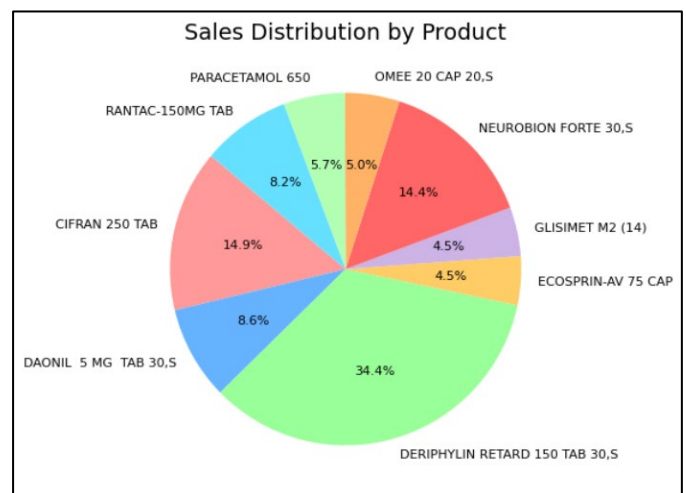


Figure 15: Box plot for quantity sold per product

And while calculating the Re-Order Point, the lead time for all products are very less. It ranges from 0 to 3 days to get restocked. So, from this can analyze that in general the pharmacy does not need to maintain large inventory. This directly reduces storage costs and holding expenses. Since lead times are short, the required safety stock levels can be kept relatively low, reducing unnecessary capital investment in inventory. The stock replenishment analysis revealed that maintaining the optimal reorder point is essential to prevent stock shortages and overstocking issues. By calculating the average lead time and safety stock for each product, it was determined that products such as DERIPHYLIN RETARD 150 TAB require a higher reorder point due to their demand and lead time variability. The calculated reorder points ensure that pharmacy operations remain uninterrupted while minimizing holding costs.

The EOQ analysis helped determine the optimal order quantities for minimizing inventory costs, including ordering and holding costs. Products such as DERIPHYLIN RETARD 150 TAB and NEUROBION FORTE 30,S demonstrated higher EOQ values, indicating their importance in the pharmacy's inventory strategy. This analysis provides actionable insights for bulk ordering decisions, especially for high-demand products, balancing procurement costs with storage expenses.

The sales distribution analysis for the filtered data, supported by pie chart (Figure-15), highlighted variations in product sales and identified outliers. Products like CIFRAN 250 TAB and DERIPHYLIN RETARD 150 TAB exhibited higher sales quantities with notable demand fluctuations, indicating their critical role in meeting customer needs. The identification of such patterns can help optimize stock allocation and reorder schedules.

In sales forecasting, time series forecasting using the ARIMA model provided reliable predictions for future sales trends. With the monthly sales trend graph for each product, the sales of DERIPHYLIN RETARD 150 TAB show a consistent decline over the observed months. This might indicate seasonal demand patterns, market competition, or changes in customer preferences. Rest maintains a relatively stable sales pattern without significant fluctuations. These products may be considered as steady-demand items. There are no evident spikes that indicate high demand during specific months, which suggests the absence of a seasonal pattern for the current observation period. Given the relatively steady nature of most products and declining sales of only one product and not much influence of seasons or unexpected trends, future sales forecasting models with past trends improve accuracy.

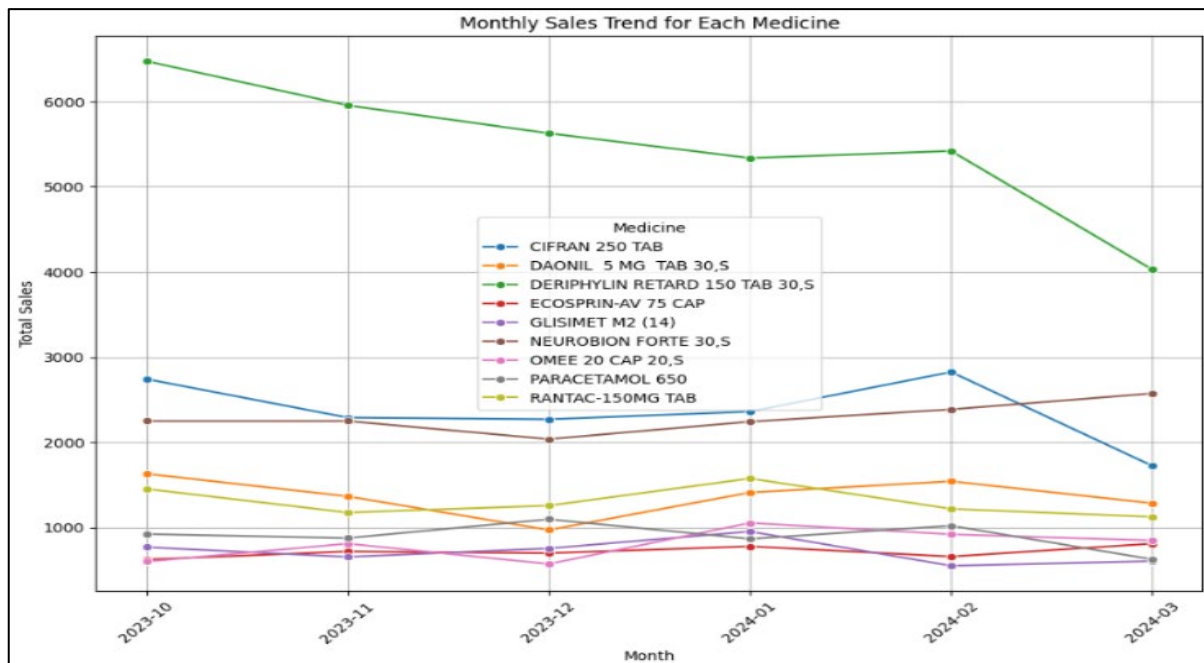


Figure 16: Line chart for monthly sales trend for each product

The forecasted result with the ARIMA is given in Figure 17,

	Product Name	April-2024	May-2024	June-2024
0	CIFRAN 250 TAB	2860.536579	2305.357633	2200.433350
1	DAONIL 5 MG TAB 30,S	1084.536235	1398.448064	1535.367335
2	DERIPHYLIN RETARD 150 TAB 30,S	4271.217797	4629.683300	4255.995710
3	ECOSPRIN-AV 75 CAP	688.282951	813.606190	712.823094
4	GLISIMET M2 (14)	924.960166	650.579908	656.128667
5	NEUROBION FORTE 30,S	2497.833681	2580.069225	2500.831638
6	OMEE 20 CAP 20,S	1133.505903	774.281994	1030.680835
7	PARACETAMOL 650	801.218958	455.497840	689.434628
8	RANTAC-150MG TAB	1569.873174	1367.645260	1061.140718

Figure 17: Forecasted results using ARIMA

This Forecasting sales trend provides the pharmacy with foresight for proactive inventory decisions, ensuring the availability of high-demand products. These findings provide a strong foundation for optimizing inventory management, ensuring better resource allocation, and improving the pharmacy's operational efficiency. Regular monitoring and updates to these models will further enhance their effectiveness as the market landscape evolves.